4. Scheduling Program

Code:

```c
#include <stdio.h>

int main()
{
    int n, i, j, temp;
    int bt[20], p[20], wt[20], tat[20];
    float avgwt = 0, avgtat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        p[i] = i + 1;
        printf("Enter burst time of P%d: ", p[i]);
        scanf("%d", &bt[i]);
    }

    // Sort processes by burst time
    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (bt[i] > bt[j])
            {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;

                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for (i = 1; i < n; i++)
        wt[i] = wt[i - 1] + bt[i - 1];

    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
        avgwt += wt[i];
        avgtat += tat[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
        printf("P%d\t%d\t%d\t%d\n",
               p[i], bt[i], wt[i], tat[i]);

    printf("\nAverage Waiting Time = %.2f", avgwt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avgtat / n);

    return 0;
}
```

Output:

```
Enter number of processes: 5
Enter burst time of P1: 6
Enter burst time of P2: 4
Enter burst time of P3: 1
Enter burst time of P4: 2
Enter burst time of P5: 5

Process BT      WT      TAT
P3      1       0       1
P4      2       1       3
P2      4       3       7
P5      5       7       12
P1      6       12      18

Average Waiting Time = 4.60
Average Turnaround Time = 8.20
```